F Extended Analysis of VQA in Text Space

We evaluate caption-based visual question answering on the MME benchmark [Fu et al., 2023a], following the "VQA without V" setup of Chan et al. [2025]. Each image in MME is first converted into a natural language caption using Qwen2.5-VL-3B-Instruct. To ensure captions remain faithful to the source modality, we use a category-specific prompting strategy during captioning: for categories involving code or mathematics, the model is instructed to *transcribe line-by-line without interpretation*; for all other categories, the model is instructed to *describe only visible characteristics without interpretation or commentary*. This prevents the captioner from injecting additional semantic reasoning that would not be directly available from the image itself.

Given these captions, we then construct (caption, Q) pairs and evaluate Qwen3-14B as the answering model under two different prompting conditions:

- No cue: `You will be given a CAPTION and a question. Your role is to answer the question only with 'yes' or 'no' by using the CAPTION. CAPTION: {caption}`
- SEE cue: `You will be given a CAPTION and a question. Your role is to answer the question only with 'yes' or 'no' by imagining what it looks like to see the CAPTION. CAPTION: {caption}`

The key manipulation is the addition of the *visual framing cue* ("imagine what it would look like to see…"), which biases the model toward perceptual simulation when interpreting the caption. Both conditions are evaluated across all categories of the MME benchmark except OCR, which inherently requires direct text recognition from images.

In the following, we present an example drawn from the MME dataset. The example shows the original image, the generated caption, the associated yes/no question, and the generations from Qwen3-14B under both cue conditions. These examples illustrate the mechanism by which sensory prompting allows the model to answer correctly by invoking a visual imagination of the scene's text caption.

F.1 Example: artwork/10256.jpg

Q: [Y] Does this artwork exist in the form of painting?

Figure 31: Caption: A winter scene with people skating on a frozen river, horses pulling carriages, and buildings in the background under a cloudy sky.

No cue output (incorrectly answers [N])

Okay, let's see. The user provided a caption describing a winter scene with people skating, horses pulling carriages, buildings under a cloudy sky. The question is whether this artwork exists as a painting. The answer needs to be yes or no based on the caption.

First, the caption itself is a description of a scene. The question is about whether there's a painting that depicts this. But the caption doesn't mention anything about a painting. It just describes the scene. So, the caption is likely a description of a painting, but the existence of the artwork as a painting isn't confirmed by the caption. The caption could be from a photograph, a drawing, or even a written description. Without explicit mention of it being a painting, I can't assume it's a painting. Therefore, the answer would be no because the caption doesn't state that it's a painting. The user might be testing if I can infer the medium from the description, but since it's not specified, the answer is no.

SEE output (correctly answers [Y])

Okay, let's see. The user provided a caption describing a winter scene with people skating, horses pulling carriages, buildings under a cloudy sky. The question is whether this artwork exists as a painting. They want a yes or no answer.

First, I need to figure out if the caption is referring to a real painting. The caption sounds like a description that could be from a well-known painting. The elements mentioned—frozen river, skaters, horse carriages, buildings, cloudy sky—are classic elements found in 19th-century landscape paintings.

Wait, the most famous painting that comes to mind is "Winter Landscape with Skaters" by Hendrick Avercamp. He was a Dutch painter known for his winter scenes. That painting does have people skating on a frozen river, horse-drawn carriages, and buildings in the background with a cloudy sky. So if the caption is describing that painting, then the answer would be yes.But how do I know if the caption is referring to that specific painting? The user didn't mention the title or artist. However, the elements are very specific and match Avercamp's work. Since the question is whether the artwork exists as a painting, and given that such a painting does exist, the answer would be yes.

Alternatively, maybe the user is asking if the described scene is a real painting, not just a hypothetical one. Since the description matches a known painting, the answer is yes. If there's no such painting, the answer would be no, but in this case, there is. So the answer should be yes.

G Full Prompted Generation Examples

All generations in this section originate from the following caption (WiT):

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Awaiting a pitch — batter, catcher, and umpire in baseball
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G.1 128-token Text Generations

Below are full 128-token text generations from Qwen3-32B when prompted with the template outlined in Figure 3.

SEE output (128 token generation)

Okay, the user wants me to imagine what it looks like to see a baseball scene where the batter, catcher, and umpire are waiting for a pitch. Let me break this down. First, I need to visualize a baseball field, specifically the area around home plate. The batter is up to bat, so they're standing in the batter's box, maybe adjusting their grip on the bat or their stance. The catcher is behind the plate, wearing all that gear—helmet, chest protector, knee pads. They're probably crouched low, watching the pitcher closely. The umpire is standing near home plate,

HEAR output (128 token generation)

Okay, the user wants me to imagine what it would sound like to hear the moment before a pitch in baseball, involving the batter, catcher, and umpire. Let me break this down. First, I need to think about the typical sounds in a baseball game during that specific moment. The atmosphere in the stadium, the crowd's reactions, the communication between players, and any equipment sounds.

Starting with the batter: they might be adjusting their batting gloves, maybe tapping the plate with their foot or the bat. There's the sound of the bat being raised, maybe some deep breathing. The batter might also be listening to the

No cue output (128 token generation)

Okay, so the user wants me to imagine a scenario in baseball where a batter, catcher, and umpire are involved, specifically awaiting a pitch. Let me break this down. First, I need to visualize the setup. The batter is at the plate, ready to hit. The catcher is behind home plate, crouched down, signaling the pitcher. The umpire is positioned behind the batter, right? Or maybe there are multiple umpires, but the main one here is the home plate umpire.

I should think about the sequence of events. The pitcher is about to throw the ball. The batter is focused,

G.2 Sensory Redirection Generations

Below are full 128-token text generations from Qwen3-32B when prompted with the sensory redirecting template outlined in Figure 7, using outputs from Appendix G.1.

SEE-to-HEAR output (128 token generation)

The scene hums with anticipation, the only sounds the soft rustle of grass underfoot and the occasional murmur of the crowd in the distance. A tense silence hangs in the air as the batter shifts their weight, the creak of leather as they adjust their grip on the bat. The catcher's gear creaks and shifts as they crouch low behind the plate, their breath steady but audible, the faint clink of their mask as they glance toward the pitcher. The umpire stands rigid, his voice low and firm as he calls the count, each word crisp and clear, cutting through the quiet. Then, a sudden

HEAR-to-SEE output (128 token generation)

The scene is tense and electric, frozen in the instant before the pitch. The batter stands at the plate, gripping the bat tightly, his muscles coiled with anticipation. His eyes are locked onto the pitcher, scanning for the slightest movement. The catcher crouches behind the plate, gloved hands poised, his body ready to react. His eyes are on the umpire, waiting for the signal that will set the play in motion. The umpire stands at the center of the action, one hand raised, his posture rigid with authority. Around them, the stadium hums with quiet intensity—fans leaning forward in their seats, the

G.3 Sensory Ablation Generations

Below are full 128-token text generations from Qwen3-32B when prompted with the sensory ablation prompt in Figure 7, using outputs from Appendix G.1.